

Multiple Choice

1. **Answer:** D

Options: A) Plants keep water in their roots.; B) Plants release nitrogen into the soil.; C) Plants provide shelter from the weather.; D) Plants make sugar in the presence of light.

Note: Plants make sugar through photosynthesis, using sunlight to convert carbon dioxide and water into glucose, which serves as a primary energy source for animals that consume them.

2. **Answer:** B

Options: A) giraffe's long neck; B) baby deer's spots; C) otter's oily fur; D) fish's gills

Note: The spots on baby deer serve as a camouflage adaptation, helping them blend into their surroundings and evade detection by predators.

3. **Answer:** D

Options: A) The planet closest to the Sun.; B) The planet with the longest day.; C) The planet with the most moons.; D) The planet farthest from the Sun.

Note: The planet farthest from the Sun takes the longest time to orbit the Sun due to its greater distance, resulting in a longer year compared to the other planets.

4. **Answer:** C

Options: A) The water became a gas.; B) The mass of water increased.; C) The water took a definite shape.; D) The flavor of the water changed.

Note: As water freezes, it transitions from a liquid state, where it takes the shape of its container, to a solid state, where it maintains a definite shape and volume.

5. **Answer:** D

Options: A) The amplitude would double.; B) The frequency would double.; C) The amplitude would decrease by half.; D) The frequency would decrease by half.

Note: The frequency of a wave is inversely related to its wavelength, so if the wavelength is doubled, the frequency decreases by half.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: An increase in the use of automobiles contributes to higher carbon dioxide levels, which are linked to global warming rather than causing unusually cold weather.

7. **Answer:** True

Options: A) True; B) False

Note: A tornado creates a narrow path of destruction due to its intense rotating winds, which can uproot trees and damage structures along a specific track as it travels through an area.

8. **Answer:** False

Options: A) True; B) False

Note: While developing multiple hypotheses can be beneficial in scientific investigations, the most crucial step is actually testing these hypotheses through experimentation and analyzing the data to draw valid conclusions.

9. **Answer:** True

Options: A) True; B) False

Note: An apple core is made of organic material that is biodegradable and can be broken down by microorganisms, while a glass bottle, metal can, and plastic bottle are non-biodegradable and take much longer to decompose.

10. **Answer:** False

Options: A) True; B) False

Note: Alternative fuels are not widely available at every gas station, as most stations primarily offer conventional gasoline and diesel, limiting drivers' access to alternative fuel options.

Long Form Response

11. **Answer:** Students can effectively measure and communicate the size of oak tree leaves by using a ruler or measuring tape to determine their length and width, ensuring they record these measurements accurately. Labeling each measurement is essential as it helps to clarify what the numbers represent, such as "length in centimeters" or "width in inches." This practice not only enhances understanding but also allows for easier comparison and analysis of data collected by different students or over time. Clear labeling ensures that information is communicated effectively, reducing the chances of confusion and promoting better scientific communication. By organizing their findings with proper labels, students can better engage in discussions about their observations and share their results with peers.

Note: correct because it emphasizes the importance of accurate measurement and clear labeling in effectively communicating data about the size of oak tree leaves, which facilitates understanding and comparative analysis in a scientific context.

12. **Answer:** To effectively display and compare the percentages of minerals found in granite samples from western Nevada, I would choose a bar graph format. This visual representation allows for easy comparison between different minerals, as each mineral can be represented by a separate bar, with the height corresponding to its percentage in the granite samples. Additionally, I would label each bar clearly with the name of the mineral and its percentage, making the data easily interpretable for viewers. This format not only enhances understanding but also engages students in analyzing the

composition of granite, fostering a deeper appreciation for geology.

Note: correct because a bar graph effectively visualizes and compares the percentages of different minerals, allowing for clear interpretation and easy identification of each mineral's proportion in the granite samples.

End of Answer Key